
MAE 5730: Intermediate Dynamics and Vibrations

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QUESTION 29: MASS IN SLOT ON TURNTABLE

TOP VIEW

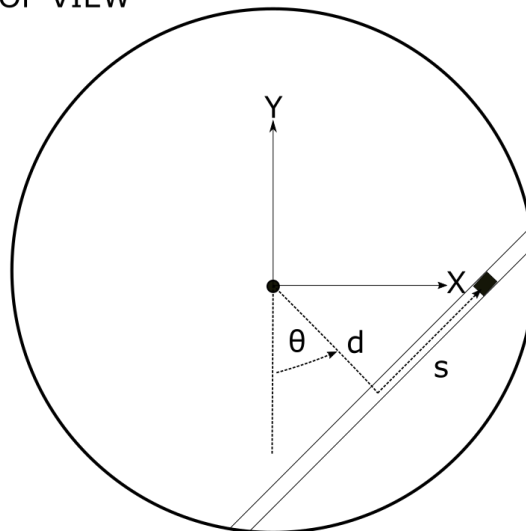


Figure 1: Mass in slot on turntable

- Rigid turntable is free to rotate about a hinge at its center (m_t, I_t)
- Straight friction-less slot passes at a distance d from center
- Mass m_s slides in the slot

A) FIND THE ACCELERATION OF THE MASS IN TERMS OF $d, \theta, \dot{\theta}, \ddot{\theta}, s, \dot{s}, \ddot{s}$

METHOD 1: USING BASE VECTORS $\hat{i}$ AND $\hat{j}$

TOP VIEW

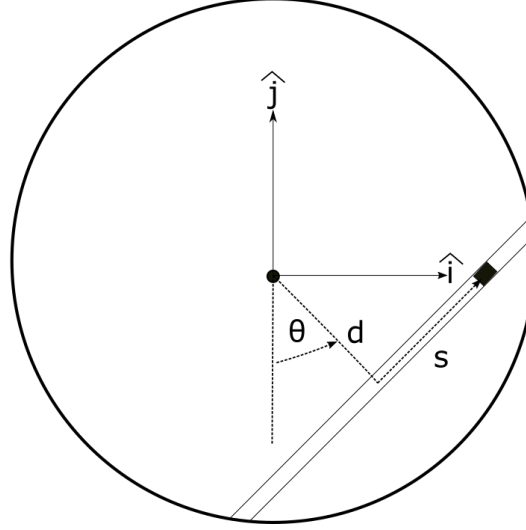


Figure 2: Using base vectors $\hat{i}$ and $\hat{j}$

Position of mass:

$$\vec{r} = (d \sin \theta + s \cos \theta) \hat{i} + (-d \cos \theta + s \sin \theta) \hat{j} \quad (1)$$

Velocity of mass:

$$\vec{v} = \frac{d\vec{r}}{dt} = (d \cos \theta \dot{\theta} + \dot{s} \cos \theta - s \sin \theta \dot{\theta}) \hat{i} + (-d \sin \theta \dot{\theta} + \dot{s} \sin \theta + s \cos \theta \dot{\theta}) \hat{j} \quad (2)$$

Acceleration of mass:

$$\begin{aligned} \vec{a} = \frac{d\vec{v}}{dt} = & (-d \sin \theta \dot{\theta}^2 + d \cos \theta \ddot{\theta} + \dot{s} \cos \theta - 2\dot{s} \sin \theta \dot{\theta} - s \cos \theta \dot{\theta}^2 - s \sin \theta \ddot{\theta}) \hat{i} \\ & + (d \cos \theta \dot{\theta}^2 + d \sin \theta \ddot{\theta} + \dot{s} \sin \theta + 2\dot{s} \cos \theta \dot{\theta} - s \sin \theta \dot{\theta}^2 + s \cos \theta \ddot{\theta}) \hat{j} \end{aligned} \quad (3)$$

METHOD 2: USING BASE VECTORS $\hat{i}'$ AND $\hat{j}'$

Use the fact that $\dot{\hat{i}}' = \vec{\omega} \times \hat{i}' = \omega \hat{j}'$ and $\dot{\hat{j}}' = -\vec{\omega} \times \hat{j}' = -\omega \hat{i}'$

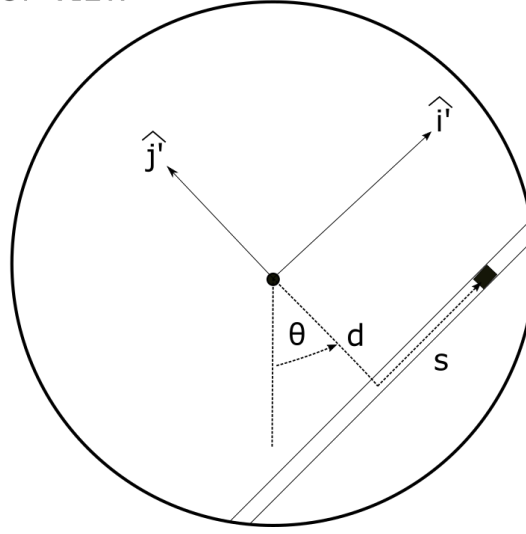
Position of mass:

$$\vec{r}' = s \hat{i}' - d \hat{j}' \quad (4)$$

Velocity of mass:

$$\begin{aligned} \vec{v}' &= \frac{d\vec{r}'}{dt} \\ &= \dot{s} \hat{i}' + s \dot{\hat{i}}' - d \dot{\hat{j}}' \\ &= (\dot{s} + d\dot{\theta}) \hat{i}' + s\dot{\theta} \hat{j}' \end{aligned} \quad (5)$$

TOP VIEW

Figure 3: Using base vectors $\hat{i}'$ and $\hat{j}'$

Acceleration of mass:

$$\begin{aligned}
 \vec{a}' &= \frac{d\vec{v}'}{dt} \\
 &= \ddot{s}\hat{i}' + \dot{s}\dot{\hat{i}}' + \dot{s}\dot{\theta}\hat{j}' + s\ddot{\theta}\hat{j}' + s\dot{\theta}\dot{\hat{j}}' + d\ddot{\theta}\hat{i}' + d\dot{\theta}\dot{\hat{i}}' \\
 &= (\ddot{\theta}d + \ddot{s} - \dot{\theta}^2 s)\hat{i}' + (2\dot{s}\dot{\theta} + s\ddot{\theta} + d\dot{\theta}^2)\hat{j}'
 \end{aligned} \tag{6}$$

METHOD 3: USING THE 5 TERM ACCELERATION FORMULA

Use $\vec{v}_{rel} = \dot{s}\hat{i}'$ and $\vec{a}_{rel} = \ddot{s}\hat{i}'$

Here,

$$\vec{a}_{O'/O} = \ddot{\theta}d\hat{i}' + \dot{\theta}^2 d\hat{j}' \tag{7}$$

$$\vec{a}_{m/O'} = \vec{a}_{rel} = \ddot{s}\hat{i}' \tag{8}$$

$$\vec{\omega} = \dot{\theta}\hat{k} \tag{9}$$

$$\vec{v}_{m/O'} = \vec{v}_{rel} = \dot{s}\hat{i}' \tag{10}$$

So we can write the acceleration of mass m using the 5 term acceleration formula:

$$\begin{aligned}
 \vec{a}_{m/O} &= \vec{a}_{O'/O} + \vec{a}_{m/O'} + \vec{\omega} \times (\vec{\omega} \times \vec{r}_{m/O'}) + \dot{\vec{\omega}} \times \vec{r}_{m/O'} + 2\vec{\omega} \times \vec{v}_{m/O'} \\
 &= -\dot{\theta}^2 d\hat{j}' + \ddot{\theta}d\hat{i}' + \ddot{s}\hat{i}' - \dot{\theta}^2 s\hat{i}' + \ddot{\theta}s\hat{j}' + 2\dot{\theta}\dot{s}\hat{j}' \\
 &= (\ddot{\theta}d + \ddot{s} - \dot{\theta}^2 s)\hat{i}' + (-\dot{\theta}^2 d + \ddot{\theta}s + 2\dot{\theta}\dot{s})\hat{j}'
 \end{aligned} \tag{11}$$

TOP VIEW

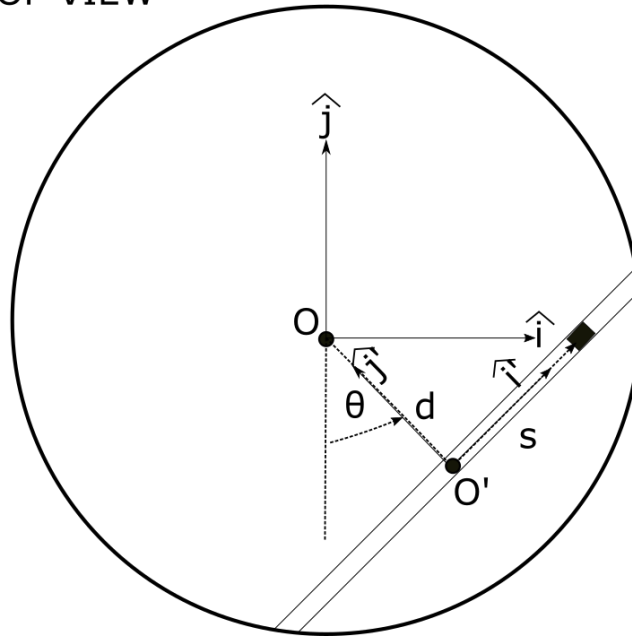


Figure 4: Using the 5 term acceleration formula

CHECK THAT ALL THREE METHODS GIVE THE SAME ANSWER

If we substitute $\hat{i} = \cos\theta \hat{i}' - \sin\theta \hat{j}'$ and $\hat{j} = \sin\theta \hat{i}' + \cos\theta \hat{j}'$ in the acceleration obtained using Method 1 i.e. Eq.3, then the acceleration of mass is given by:

$$\vec{a} = (\ddot{\theta}d + \ddot{s} - \dot{\theta}^2 s)\hat{i}' + (2\dot{s}\dot{\theta} + s\ddot{\theta} + d\dot{\theta}^2)\hat{j}' \quad (12)$$

This is the same as acceleration of mass obtained using Method 2 (Eq.6) and Method 3 (Eq.11). Thus all the three methods give the same answer.

B) FIND THE EQUATIONS OF MOTION USING AMB FOR THE WHOLE SYSTEM ABOUT THE CENTER AND LMB FOR THE MASS

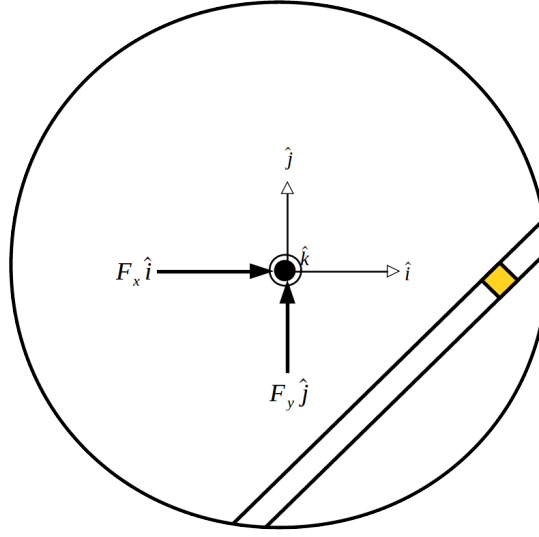


Figure 5: FBD of 'turntable+mass' system

As seen from the FBD in Fig.5, there is no torque about the axis perpendicular to the plane and passing through the center of turntable. Thus Angular Momentum Balance about this axis gives:

$$\begin{aligned} \sum \vec{r} \times m_i \vec{a}_i &= 0 \\ \therefore I_t \ddot{\theta} \hat{k} + \vec{r}_m \times m \vec{a}_m &= 0 \end{aligned} \quad (13)$$

Substituting the values of $\vec{r}_m$ and $\vec{a}_m$ from Eq.4 and Eq.6 respectively, and taking dot product with $\hat{k}$ we get:

$$\left(\frac{I_t}{m} + d^2\right)\ddot{\theta} + d\ddot{s} + s^2\ddot{\theta} + 2s\dot{s}\dot{\theta} = 0 \quad (14)$$

Using the FBD shown in Fig.6, Linear Momentum Balance for mass 'm':

$$\begin{aligned} \sum \vec{F} &= m \vec{a} \\ \therefore F_n \hat{j}' &= m[(\ddot{\theta}d + \ddot{s} - \dot{\theta}^2 s)\hat{i}' + (\dot{\theta}^2 d + 2\dot{s}\dot{\theta} + s\ddot{\theta})\hat{j}'] \end{aligned} \quad (15)$$

Taking dot product with $\hat{i}'$

$$\ddot{\theta}d + \ddot{s} - \dot{\theta}^2 s = 0 \quad (16)$$

Solving Eq.14 and Eq.15,

$$\boxed{\ddot{\theta} = \frac{-ms\dot{\theta}(2\dot{s} + d\dot{\theta})}{I_t + ms^2}} \quad (17)$$

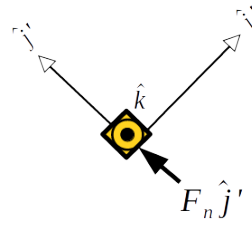


Figure 6: FBD of mass

$$\ddot{s} = \frac{s\dot{\theta}(I_t\dot{\theta} + d^2m\dot{\theta} + ms^2\dot{\theta} + 2dm\dot{s})}{I_t + ms^2} \quad (18)$$

C) FIND THE EQUATIONS OF MOTION USING LAGRANGE EQUATIONS AND CHECK FOR AGREEMENT.

Generalized co-ordinates for Lagrange Equations: θ, s

Define kinetic energy of system:

$$E_k = \frac{1}{2}mv_m^2 + \frac{1}{2}I_t\dot{\theta}^2 \quad (19)$$

Define potential energy of system:

$$E_p = 0 \quad (20)$$

Lagrange's method:

$$\mathcal{L} = E_k - E_p = \frac{1}{2}mv_m^2 + \frac{1}{2}I_t\dot{\theta}^2 \quad (21)$$

The equations of motion are obtained using:

$$\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = 0 \quad (22)$$

$$\ddot{\theta} = \frac{-ms\dot{\theta}(2\dot{s} + d\dot{\theta})}{I_t + ms^2} \quad (23)$$

$$\ddot{s} = \frac{s\dot{\theta}(I_t\dot{\theta} + d^2m\dot{\theta} + ms^2\dot{\theta} + 2dm\dot{s})}{I_t + ms^2} \quad (24)$$

These equations are same as Eq.17 and Eq.18 which were obtained using AMB/LMB .

D) ASSUME ICs THAT $s(0) = 0$, $\dot{s}(0) = v_0$, $\theta(0) = 0$ AND $\dot{\theta}(0) = \omega_0 > 0$.
 AS $t \rightarrow \infty$ DOES $\theta \rightarrow \infty$?

As seen from Eq.23 and Eq.24, the equations of motion of this system contain the system parameters m , I_t and d . The actual motion also depends upon the initial conditions. There is no explicit solution for these coupled differential equations.

For the given ICs, there are 2 special cases:

- When $m = 0 \Rightarrow \theta = \omega_0 t \Rightarrow$ As $t \rightarrow \infty$, $\theta \rightarrow \infty$
- When $\dot{s}(0) = v_0 = 0$, $\Rightarrow \theta = \omega_0 t \Rightarrow$ As $t \rightarrow \infty$, $\theta \rightarrow \infty$

For $m \neq 0$ and $v_0 \neq 0$:

Consider a case where ICs are: $s(0) = 0$, $\dot{s}(0) = v_0 = 1$, $\theta(0) = 0$ and $\dot{\theta}(0) = \omega_0 = 1$.

System parameters are: $m = 1$, $I_t = 1$ and $d = 1$

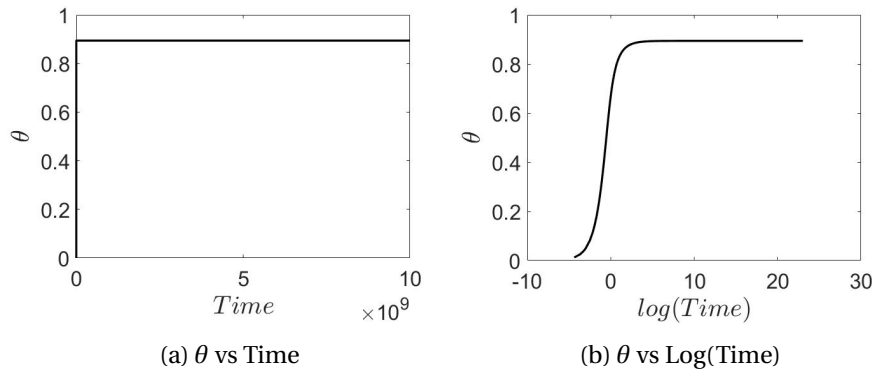


Figure 7: Numerical Solution

From Fig.7a and Fig.7b, it looks like the value of θ is bounded above, even if we plot it as a function of $\log(t)$. So we can almost certainly (with confidence $\rightarrow 90\%$:)) conclude that as $t \rightarrow \infty$, θ **does not** go to infinity.

To analytically prove that as $t \rightarrow \infty$, θ does not go to infinity, we need to prove that $\lim_{t \rightarrow \infty} \theta(t) = A$ finite real number! (For $m \neq 0$ and $v_0 \neq 0$)

Let us begin by looking at the Angular Momentum Balance of the entire system. From Fig.5, since there is no torque about the out-of-plane axis passing through center, the angular momentum of the system about this axis is conserved. Thus,

$$I_t \dot{\theta} \hat{k} + m \vec{r}_m \times \vec{v}_m = I_t \dot{\theta}(0) \hat{k} + \vec{r}_m(0) m \times \vec{v}_m(0)$$

$$\therefore (I_t + md^2 + ms^2) \dot{\theta} + md \dot{s} = (I_t + md^2 + ms(0)^2) \dot{\theta}(0) + md \dot{s}(0) = C_0$$

$$\therefore \dot{\theta} = \frac{C_0 - md \dot{s}}{I_t + md^2 + ms^2} \quad (25)$$

where C_0 is the initial angular momentum of the system. For the ICs given in question,

$$C_0 = (md^2 + I_t)\omega_0 + mdv_0 \quad (26)$$

For simplicity, assume that $\omega_0 > 0$ and $v_0 > 0 \implies C_0 > 0$

From the equations of motion in Eq.24, we see that $\ddot{s} > 0$ for $[s, \dot{\theta}, \dot{s}] > 0$. This can be interpreted as follows: As the mass starts out to move away from the center, its velocity increases. In the process it takes away the energy of turntable. But since the total energy of the system has to be conserved (no external forces that do work), the velocity of mass is bounded above. (Similarly $\dot{\theta}$ is also bounded above). So $\lim_{t \rightarrow \infty} \dot{s} = C_1 \implies s(t) \approx C_1 t$ (for large t) where C_1 is positive constant that depends on the initial energy of the system. Substituting this in Eq.25,

$$\dot{\theta} \approx \frac{C_0 - mdC_1}{I_t + md^2 + mC_1^2 t^2} \quad (27)$$

Thus $\dot{\theta}$ has the form of $\frac{A}{B+Ct^2}$ where A,B,C are some finite real constants. This implies that θ will be of the form: $\tan^{-1}(E)$ which is bounded! Therefore θ does not go to infinity as $t \rightarrow \infty$ except for $m = 0$ or $v_0 = 0$.

[Just for the sake of completeness, here's the actual expression for θ :

$$\begin{aligned} \therefore \theta &= \int \frac{C_0 - mdC_1}{I_t + md^2 + mC_1^2 t^2} dt \\ &= \frac{(C_0 - mdC_1) \tan^{-1} \frac{C_1 \sqrt{m} t}{\sqrt{d^2 m + I_t}}}{C_1 \sqrt{m} \sqrt{d^2 m + I_t}} + C_3 \\ \therefore \lim_{t \rightarrow \infty} \theta &= \frac{(C_0 - mdC_1)}{C_1 \sqrt{m} \sqrt{d^2 m + I_t}} \cdot \frac{\pi}{2} + C_3 \end{aligned} \quad (28)$$

Since C_1, m are positive finite real numbers, we conclude that θ is bounded as $t \rightarrow \infty$.]

```

%Author Alap Kshirsagar
close all; clear all;
%Root file for mass in slot problem
% Apart from this main file there are 3 other functions used in this code:
% 2) slot_EOM solves AMB/LMB equations symbolically and writes into
% slot_rhs file which is used by ODE45 in this file
% 3) slot_lagrange obtains equations of motion using Lagrange's method and
% writes into slot_lagrange_rhs which is used by ODE45 in this file

%Define timespan of simulation
timespan = 10^10;

%Define parameters
p.m = 1; %Mass of mass in slot
p.mt = 1; %Mass of turntable
p.It = 1; %Moment of inertia of turntable
p.d = 1; %Shortest distance between slot and center of table
p.g = 1; %Acc. due to gravity

%Define initial conditions
s_init = 0;
sdot_init = 1;
theta_init = 0;
thetadot_init = 1;

%Options for ODE45
opts.RelTol = 1e-6;
opts.AbsTol = 1e-6;

%Generate RHS file with equations of motion
slot_EOM();
slot_lagrange();

%Solve ODEs
[time,result] = ode45(@slot_rhs,[0 timespan],[theta_init,thetadot_init,s_init,
sdot_init],opts,p);
[timeLagrange,resultLagrange] = ode45(@slot_lagrange_rhs,[0 timespan],[theta_init,
thetadot_init,s_init,sdot_init],opts,p);

%Position of mass
position_mass = [p.d*sin(result(:,1))+result(:,3).*cos(result(:,1)), -p.d*cos(result
(:,1))+result(:,3).*sin(result(:,1))];
position_mass_lagrange = [p.d*sin(resultLagrange(:,1))+resultLagrange(:,3).*cos
(resultLagrange(:,1)), -p.d*cos(resultLagrange(:,1))+resultLagrange(:,3).*sin
(resultLagrange(:,1))];

%Thetadoubledot
thetadoubledot = -(p.m*result(:,3).*result(:,2).*(2*result(:,4) + p.d*result(:,3)))./
(p.It + p.m*result(:,3).^2);
thetadoubledotLagrange = -(p.m*resultLagrange(:,3).*resultLagrange(:,2).*(
2*resultLagrange(:,4) + p.d*resultLagrange(:,3)))./(p.It + p.m*resultLagrange(:,3).
^2);

```

```

%Plot results
plot(timeLagrange,resultLagrange(:,1), 'k', 'LineWidth',2);
ylabel('$\theta$', 'Interpreter', 'latex', 'FontSize',24);
xlabel('$Time$', 'Interpreter', 'latex', 'FontSize',24);

figure;
plot(timeLagrange,log(resultLagrange(:,1)), 'k', 'LineWidth',2);
ylabel('$\log(\theta)$', 'Interpreter', 'latex', 'FontSize',24);
xlabel('$Time$', 'Interpreter', 'latex', 'FontSize',24);

%Compare Lagrange and AMB/LMB solutions
figure;
plot(position_mass_lagrange(:,1),position_mass_lagrange(:,2), 'k.', 'LineWidth',2);
hold on;
plot(position_mass(:,1),position_mass(:,2), 'k--', 'LineWidth',2);
axis 'equal'
ylabel('$y_m$', 'Interpreter', 'latex', 'FontSize',24);
xlabel('$x_m$', 'Interpreter', 'latex', 'FontSize',24);
legend('Lagrange', 'AMB/LMB');
hold off;

```

```

function [] = slot_EOM()
%Author Alap Kshirsagar
%slot_EOM This file derives the equations of motion using
%symbolic computations and writes into a RHS file

syms theta s thetadot sdot thetadoubledot sdoubledot real
syms m mt It d g real

%Define unit vectors
i = [1;0;0];
j = [0;1;0];
k = [0;0;1];

%Define position vector of mass
r_m = (d*sin(theta)+s*cos(theta))*i+(-d*cos(theta)+s*sin(theta))*j;
%r_m = s*i-d*j;

%Define velocity of mass
v_m = i*(d*cos(theta)*thetadot+sdot*cos(theta)-s*sin(theta)*thetadot) ...
      +j*(d*sin(theta)*thetadot+sdot*sin(theta)+s*cos(theta)*thetadot);

%Define acceleration of mass
a_m = i*(-d*sin(theta)*thetadot^2 + d*cos(theta)*thetadoubledot + sdoubledot*cos(theta) - 2*sdot*sin(theta)*thetadot - s*cos(theta)*thetadot^2 - s*sin(theta)*thetadoubledot) ...
      + j*(d*cos(theta)*thetadot^2 + d*sin(theta)*thetadoubledot + sdoubledot*sin(theta) + 2*sdot*cos(theta)*thetadot - s*sin(theta)*thetadot^2 + s*cos(theta)*thetadoubledot);
%a_m = i*(thetadoubledot*d+sdoubledot-thetadot^2*s)...
%      + j*(2*sdot*thetadot+s*thetadoubledot+d*thetadot^2);

%Define Hdot (rate of change of angular momentum) for entire system about
%center

HdotMwrtO = m*cross(r_m,a_m);
HdotTable = It*thetadoubledot*k;
HdotTotal = HdotMwrtO + HdotTable;
HMwrtO = simplify(m*cross(r_m,v_m));

%Moments of forces
MaboutO = 0;

%Equations of motion
eqn1 = simplify(MaboutO-HdotTotal);
eqn2 = thetadoubledot*d+sdoubledot-thetadot^2*s;

%Solve these equations symbolically to get thetadoubledot and
%sdoubledot
[t1,t2] = solve([eqn1,eqn2],[thetadoubledot,sdoubledot]);
t1char = char(simplify(t1));
t2char = char(simplify(t2));

```

[illegible]

```

function [] = slot_lagrange()
%Author Alap Kshirsagar
%slot_lagrange This file derives the equations of motion using
%lagrange method and writes into a RHS file

syms theta s thetadot sdot thetadoubledot sdoubledot real
syms m mt It d g real

%Define unit vectors
i = [1;0;0];
j = [0;1;0];
k = [0;0;1];

%Define generalized coordinates
q = [theta s];
qdot = [thetadot sdot];
qdoubledot = [thetadoubledot sdoubledot];

%Define position vector of mass
r_m = (d*sin(theta)+s*cos(theta))*i+(-d*cos(theta)+s*sin(theta))*j;

%Define velocity of mass
v_m = i*(d*cos(theta)*thetadot+sdot*cos(theta)-s*sin(theta)*thetadot) ...
      +j*(d*sin(theta)*thetadot+sdot*sin(theta)+s*cos(theta)*thetadot);
%v_m = (sdot+d*thetadot)*i+s*thetadot*j;

%Define kinetic energy of system
KE_m = 0.5*m*(v_m'*v_m);
KE_t = 0.5*It*thetadot^2;
KE = KE_m+KE_t;

%Define potential energy of system
PE = 0;

%Lagrange method
Lag = KE-PE;
EoM = jacobian(Lag, q')' - jacobian(jacobian(Lag,qdot)',[q qdot])*[qdot
qdoubledot]';

%Simplify these equations for writing into RHS file for ODE45
M = simplify(jacobian(EoM, qdoubledot)); % mass 'matrix'
rhs_stripped = simplify(EoM - M*qdoubledot'); % for use in ODE45 rhs function
rhs = -M\rhs_stripped;

t1char = char(rhs(1));
t2char = char(rhs(2));

%Put derived equations in a right-hand-side file ode45.
fid=fopen('slot_lagrange_rhs.m','w');
fprintf(fid, 'function zdot=slot_lagrange_rhs(t,z,p) \n');
fprintf(fid, 'm = p.m ; \n');
fprintf(fid, 'mt = p.mt ; \n');
fprintf(fid, 'It = p.It ; \n');
fprintf(fid, 'd = p.d ; \n');

```

[illegible]